John ALLOU

Senior Software Engineer

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SKILLS

Programming Languages:
- Java
- TypeScript
- Python
- JavaScript
- Dart
- HTML / CSS

Frontend & Mobile Development:
- React
- Angular
- Flutter
- React Native
- Android (Java)

Cloud & DevOps:
- AWS / Azure / Heroku
- Kubernetes
- Docker
- GitHub / Bitbucket / GithubAction
- GitLab / Jenkins / CircleCI / CodeMagic / Nexus
- CI/CD pipelines

Backend Development:
- Kafka
- Spring Boot
- RabbitMQ
- Nest.js
- FastAPI
- ExpressJs
- Node.js
-Elasticsearch
- Flask
- Maven
- JUnit

Databases:
- SQL : PostgreSQL, Oracle, MySQL, MsSQL, SqlAlchimy
- NoSQL : MongoDB, DynamoDB
- Firebase
- Spark

Tools & Environments:
- Linux / Windows / Debian
- Agile (Scrum, Kanban)
- Jira, Confluence
- Communication & Leadership

Software Architecture & Design:
- API REST / WebSocket
- SOLID / Clean code
- Microservices
- DevOps & CI/CD
- Scalable architecture

EXPERIENCES

Senior Product Engineer

Doxomed, *06/2026 - Present* *Switzerland*

Swiss digital health platform connecting patients and healthcare professionals.

Designed and developed **core product features** for a fast-growing Swiss digital health platform used in **more than 40 medical institutions**, combining product conception, full-stack engineering and technical architecture. Worked across patient and professional workflows, medical records, clinical documentation and secure communication, translating healthcare requirements into coherent and maintainable product capabilities.

Played an active role in shaping technical direction, designing **modular services**, integrating **medical interoperability standards** and improving the reliability of sensitive operations. This work also includes cryptographic document signing and AI-assisted consultation transcription. My contribution extends beyond implementation, combining product understanding, architecture, security and complex integration work to help the team deliver core capabilities on foundations that support the platform's continued growth.

Senior Software Engineer

United Nations - WIPO, *06/2025 - 12/2025* *Lausanne*

Specialized Agency of the United Nations responsible for promoting innovation and intellectual property protection worlwide.

Led the redesign of the **WIPO Lex** architecture and platform, a large scale global database for intellectual property laws and treaties, with a strong focus on modernization, performance, and long term scalability. Working across the full stack with **Java, Spring Boot, Angular, TypeScript, AWS, and MySQL**, I drove key technical changes that improved both the structure of the platform and its operational efficiency.

On the backend, I reworked the application into more modular services to improve separation of concerns, maintainability, and evolution capacity. I implemented caching mechanisms, optimized database queries, and introduced asynchronous processing to reduce latency and improve responsiveness on high volume operations. On the frontend, I developed reusable Angular components and improved rendering logic to make the interface faster, more consistent, and easier to maintain over time.

I also strengthened the delivery process by improving CI/CD pipelines and contributed to the **SEP** project, where I helped streamline quality control workflows and traceability. My contribution was not limited to implementation, as I played an active role in shaping technical direction, identifying performance bottlenecks, and putting in place solutions that improved load times, system reliability, maintainability, and the overall robustness of the platform.

ELCA, *12/2022 - 03/2025* *Lausanne*

Swiss company specializing in IT consulting, software development and digital transformation.

Worked across several strategic projects in **banking, healthcare, digital platforms, and public sector environments**, combining hands on engineering with technical design, integration work, and coordination responsibilities. My contribution consistently focused on building reliable systems, improving technical quality, and helping projects move forward in a structured and scalable way.

- **Gesundheitsdatenraum Web Portal :** Acted as **Lead Developer** on the project, with responsibility across both frontend and backend development. I led API integrations, performed code reviews, onboarded new developers, and helped structure the team's technical work. Beyond delivery, I designed the technical architecture of the application and secured it using **OAuth2 / OIDC**, ensuring that authentication and access management were implemented in a robust and maintainable way. I also introduced DevOps practices to improve quality and performance, which helped reinforce delivery reliability and overall engineering discipline on the project.

  My added value on this assignment was my ability to combine architecture, implementation, security, and team enablement, which allowed the project to move with stronger technical consistency and better execution quality.

- **BCVS :** Developed backend services and integrated high performance APIs for internal banking tools used across **34 branches**, supporting thousands of daily transactions. This required building reliable integration points, ensuring interoperability with external services, and working in a context where stability and operational continuity were critical.

  My contribution went beyond pure backend delivery, as I helped make the system more robust and scalable by handling integrations carefully and aligning implementation with real client requirements. This experience demonstrates my ability to work on business critical banking systems where backend quality, integration reliability, and technical rigor directly affect day to day operations.
- **BCV :** Contributed to the design and development of an Android banking application used by more than **100,000 users**, with features such as banking transactions, QR code payments, an interactive chatbot, and customizable dynamic components. This was a large scale user facing product where reliability, usability, and quality of execution were essential.

  My contribution was valuable because I worked on advanced product features in a banking environment while collaborating with international teams to support timely and coordinated delivery. This shows my ability to deliver in complex product contexts where technical implementation must align with both business expectations and operational constraints.

- **Loterie Romande :** Developed interactive frontend features and animations for an online gaming platform used by thousands of daily active users. I translated detailed mockups and specifications into production ready interfaces, with attention to precision, responsiveness, and user experience. This work highlights my ability to combine technical implementation with execution quality on user facing systems, and to deliver polished frontend features that contribute directly to product experience.

- **Abilis :** Contributed to the continuous improvement of an e health platform used by more than **500 partner pharmacies across Switzerland**, focusing on frontend development in **React**. I also coordinated with an offshore team to maintain delivery quality and efficiency. My contribution on this project was not only technical but also collaborative, as I helped sustain a good level of engineering quality in a distributed setup. This demonstrates my ability to remain effective in multi team environments while continuing to deliver maintainable product improvements.

- **Edulog :**Worked on the redesign of the user interface and on improvements to back office features, including user management and authentication related capabilities. I also resolved critical bugs and reinforced application security. My added value here was my ability to intervene on both usability and system reliability, improving the product from a functional and technical perspective at the same time.

- **UNIGE :** Contributed to the redesign of the University of Geneva website and implemented dynamic functionalities tailored to the institution's needs, with a focus on usability and user experience. This project further demonstrated my ability to adapt implementation to specific functional requirements while keeping the delivery clean and effective.

Analyst Consultant

Axileo, *09/2021 - 11/2022* *Paris*

Engineering and technology consulting firm, offering technical expertise and support for innovative projects.

Worked on frontend and backend development, API integration, deployment, and technical documentation for telecom related platforms. Contributed to the **MSCA** project, used to manage, monitor, and create commercial offers for Orange, Bouygues Telecom, and Free, and to the **CCA** project, a customer service intranet used to manage subscriptions and customer billing.

This experience strengthened my ability to work on enterprise systems with several moving parts, where integration quality, delivery discipline, and technical clarity were important. My contribution was valuable because I was involved across several layers of the project lifecycle, from implementation to deployment and documentation, which helped support reliable delivery in operational environments.

Software Developer Anti-fraud and Terrorism

Société Générale *09/2019 - 09/2021* *Paris*

International bank offering financial services for individuals and businesses.

Worked on anti money laundering and anti terrorism systems in an international banking environment, contributing to frontend and backend development, API implementation, integration, testing, and deployment. I also contributed to **FILA e**, a monitoring and statistical analysis tool for bank fraud used internationally, and was involved in functionality planning as well as infrastructure management of the deployed server environment.

This experience is a strong illustration of my technical value in regulated contexts. It required reliability, traceability, and disciplined execution on sensitive banking tools. My contribution was valuable because I was able to support both implementation and operational aspects of systems used in compliance driven workflows, where technical mistakes have direct business and control impact.

FullStack Developer

Freelancer, *10/2019 - 12/2022* *Paris*

Tailor made project : Design and development of web and mobile applications tailored to customers' specific needs.

Project management: Complete project lifecycle management, from needs analysis to final delivery, ensuring deadlines and quality are met.

Technical documentation: Drafting of detailed technical documents, including functional specifications, user guides, and architecture documents, in response to calls for tender.

Collaboration with customers: Working closely with stakeholders to ensure a precise understanding of requirements and to provide appropriate solutions.

Maintenance and support: Updating existing applications, correcting bugs and optimizing performance.

Internship : Developer, Digital Assistant

Caisse d'allocation Familiale, *05/2019 - 08/2019* *Paris*

French public body managing social benefits to support families.

Vacation registration tool : Creation of a software solution enabling beneficiaries to easily register their children for vacation stays, with management of personal data and automation of registration processes.

Fraud tracking and payment management: Design and implementation of a digital assistant for generating fraud monitoring reports and processing payment orders on national platforms, guaranteeing data reliability and regulatory compliance.

PROJECTS / FREELANCE

Fleury - Software Exploration and Test Intelligence

- Designed and developed a software-quality platform that helps engineering teams explore application behavior, assess test coverage and investigate failures. Conceived the **product architecture and behavioral-analysis algorithms**, connecting source code, automated tests and execution evidence.

  Integrated AI-assisted analysis and testing recommendations, supported by **distributed execution**, isolated environments and recovery mechanisms. Also developed enterprise access controls and cloud/self-hosted delivery capabilities, addressing both the complexity of engineering workflows and customer deployment requirements.
- Technology: Go, Rust, React, Next.js, TypeScript, PostgreSQL, Temporal, Kafka, gRPC, Docker, Kubernetes, Git.

Sibyl - Market Intelligence & Monitoring

- Designed and developed a market-intelligence platform that brings together research, market analysis and continuous monitoring to support strategic decisions. The product connects fragmented information into a shared analytical framework, allowing users to compare opportunities, understand market dynamics and track their evolution over time.

  Conceived the **analytical model, scoring and signal-detection algorithms**, complemented by AI-assisted semantic analysis to contextualize collected information. Structured the processing architecture to support recurring analysis, historical comparison and responsive dashboards, while controlling redundant computation and processing costs. Also integrated collaboration, subscriptions and automated delivery into the commercial product.
- Technology: React, Next.js, TypeScript, NestJS, PostgreSQL, Redis, BullMQ, Docker, GitHub Actions.

Shortlistable - AI CV Optimizer

- Designed and developed a platform that helps candidates improve their CV and tailor it to job opportunities while preserving their experience and presentation. Connected CV analysis, job-fit assessment and document editing within a complete workflow, turning identified gaps into actionable recommendations.

  Conceived the **matching logic and document-engineering architecture**, combining semantic analysis, structured experience data and layout reconstruction to preserve both relevance and visual quality. Built the interactive editing and export experience, alongside subscriptions, credits and asynchronous processing, connecting product usability with the operational requirements of a commercial platform.
- Technology: React, Next.js, TypeScript, Python, FastAPI, PostgreSQL, Redis, RQ, OCR, Docker, GitHub Actions.

MyPug (03/2022 - 05/2022)

- Developed a photo-sharing social network inspired by Pinterest and Instagram. The app allows users to upload images, follow profiles, and browse a personalized feed based on preferences. Implemented real-time notifications, chat messaging, a recommendation algorithm, and a scalable architecture.
- Technology: Flutter, NodeJS, Express, Typescript, MongoDB, AWS, Docker, Socket, Firebase, Code Magic, GIT

Bloden (10/2021 - 01/2022)

- Creation of a mobile application dedicated to real-time betting and chat games, with interactive features to engage users.
- Technology: Flutter, NodeJS, Express, Typescript, MongoDB, Socket, Firebase, Code Magic, GIT

Tootsweet Dashboard (04/2021 - 07/2021)

- Design of a statistical data analysis application for a B2C discount system, aimed at optimizing offers and discounts for Tootsweet SAS users.
- Technology: React Native, BitBucket, HapiJs, Typescript, Jasmin, MongoDB, GIT.

YoutubeScrapper (05/2020 - 06/2020)

- Mobile application that analyzes YouTube video content and provides recommendations based on similar videos.
- Technology: Flutter,J2EE, CodeMagic, MySQL, GitLab.

Robot Racer (02/2019 - 05/2019)

- Development of a Raspberry Pi-based robot capable of navigating in an environment by following an established path thanks to the processing of images perceived by its camera.
- Technology: Python, SVN, Machine Learning.

DEGREES

Master's Degree Software Architecture *Paris*

ESGI

09/2019 - 07/2021

Bachelor's Degree Computer Science (MIAGE) *Paris*

Paris Descartes University

09/2018 - 05/2019

LANGUAGES

French
Native speaker

English
Fluent